Chapter 3 — Storage and Retrieval

Book: Chapter 3, pp. 69–110

Overview

A database's core jobs: store data safely, and give it back when asked. Chapter 2 took the application developer's view (data models, query languages); this chapter looks *inside*: how does a **storage engine** (the part that lays bytes out on disk or in memory and finds them again) work?

Why care if you will never write one? You still have to *pick* and *tune* one, which needs a rough model of what it does. Engines for **transactional** workloads (many small user reads/writes) behave very differently from engines for **analytics** (a few huge scans).

Three parts:

1. **Indexes for OLTP engines** (online transaction processing, defined below): *log-structured* engines (append-only files, e.g. LSM-trees) and *page-oriented* engines (update-in-place, e.g. B-trees), plus other index kinds and in-memory databases.
2. **OLTP vs analytics**: why companies build separate *data warehouses*, and star/snowflake schemas.
3. **Column-oriented storage**: storing by column, compression, sorting, writing, precomputed aggregates.

Key terms

Term	Plain-English meaning
Storage engine	Component that stores data on disk/memory and retrieves it.
Log	An append-only sequence of records (not necessarily human-readable).
Index	Extra structure derived from the main data to find records quickly. Speeds reads, slows writes.
Key-value store	Maps a key to a value, like a hash map.
Hash index	In-memory hash map from each key to the **byte offset** (position in the file) of its value.
Segment	One chunk of the log; a file closed once it reaches a certain size.
Compaction / merging	Rewriting a segment keeping only the latest value per key / combining segments into one new one.
Tombstone	Special record meaning "this key was deleted".
Checksum	Small value computed from data to detect corruption (e.g. half-written records).
SSTable (Sorted String Table)	Segment file with key-value pairs sorted by key, each key once.
Memtable	In-memory balanced tree (e.g. red-black) of recent writes, sorted by key.
LSM-tree (Log-Structured Merge-Tree)	Memtable plus a cascade of SSTables merged in the background.
Bloom filter	Memory-cheap structure that can say "this key is definitely not here".
Size-tiered / leveled compaction	Two strategies for choosing when and which SSTables to merge.
B-tree	Most common index: a balanced tree of fixed-size **pages** (usually 4 KB) updated in place. **Leaf pages** hold the keys; **branching factor** = child references per page (typically several hundred).
Write-ahead log (WAL) / redo log	Append-only file recording each B-tree change before pages are modified, for crash recovery.
Latch	Lightweight lock protecting a data structure from concurrent threads.
Copy-on-write	Write a modified copy of a page elsewhere and update parents, instead of overwriting.
B+ tree	B-tree variant whose interior pages store abbreviated keys.
Write amplification	One logical write causing multiple physical disk writes over time.
Fragmentation	Unused gaps inside files/pages.
Primary key / secondary index	Uniquely identifies a row/document/vertex / index on a non-unique field.
Heap file	Rows stored in no particular order; indexes point into it.
Forwarding pointer	Note at a record's old heap location pointing to its new one.
Clustered / covering index	Stores the whole row in the index / stores some columns (included columns) so some queries need only the index.
Concatenated / multi-dimensional index	Several columns joined into one key in fixed order / filters on several columns at once (e.g. lat *and* lon).

Term	Plain-English meaning
R-tree / space-filling curve	Specialised spatial index / maps multi-D points to one number for a normal B-tree.
Edit distance / Levenshtein automaton	Number of single-letter insertions, deletions, substitutions between words / state machine finding words within a given edit distance.
In-memory database	Serves all reads from RAM (may log to disk for durability).
Anti-caching	Evicting least-recently-used records from an in-memory DB to disk, reloading on access.
NVM (non-volatile memory)	Memory that keeps contents without power.
Transaction (here)	Group of reads and writes as one logical unit (need not be ACID).
OLTP / OLAP	Online transaction / analytic processing: many small key-based user reads and writes / few queries scanning huge numbers of rows for aggregates.
Business intelligence	Reports and analyses that help management decide.
Data warehouse / ETL	Separate read-only-copy database for analysts / Extract–Transform–Load process feeding it.
Star schema (dimensional modeling) / snowflake schema	Central **fact table** (one row per event) surrounded by **dimension tables** (who/what/where/when/how/why) / dimensions split into sub-dimensions.
Row- / column-oriented storage	All values of a row adjacent / all values of a column together, each column in its own file.
Bitmap / run-length encoding	Per distinct value, one bit per row / store "value, count" runs.
Vectorized processing / SIMD	Tight loops over compressed column chunks in CPU cache / CPU instructions processing many values at once.
Sort key	Column(s) by which a column store orders rows.
Materialized / virtual view	Stored copy of query results kept up to date / saved query expanded on each use.
Data cube / OLAP cube	Grid of precomputed aggregates grouped by several dimensions.

Data Structures That Power Your Database

The world's simplest database. The book builds a key-value store from two **Bash** functions (`db_set` , `db_get`) storing JSON documents about cities. Our equivalent:

```
#!/bin/bash
put () {                        # usage: put <key> <value>
  echo "$1,$2" >> store.log     # append "key,value" as a new line at the end of the file
}
fetch () {                      # usage: fetch <key>
  grep "^$1," store.log |       # find every line whose key matches
    sed -e "s/^$1,//" |         # strip the "key," prefix, leaving only the value
    tail -n 1                   # keep only the LAST match = the newest value
}
```

`put` never edits old lines, so all versions of a key stay; `fetch` takes the **last** match because the newest write wins (`put 7 '{"city":"Oslo"}'` then `put 7 '{"city":"Bergen"}'` → `fetch 7` returns Bergen).

- **Writes are fast:** appending is about the cheapest disk operation, and many real databases use exactly this internally: a **log**.
- **Reads are slow:** `fetch` scans the whole file, **O(n)** — double the records, double the lookup time.

> **"Log" in this book** means any *append-only sequence of records*, possibly binary, not debug text output.

Real databases also handle concurrency, stopping the log growing forever, and crashes leaving half-written records — but the principle stays.

Indexes. Extra metadata acting as a signpost to find keys without scanning:

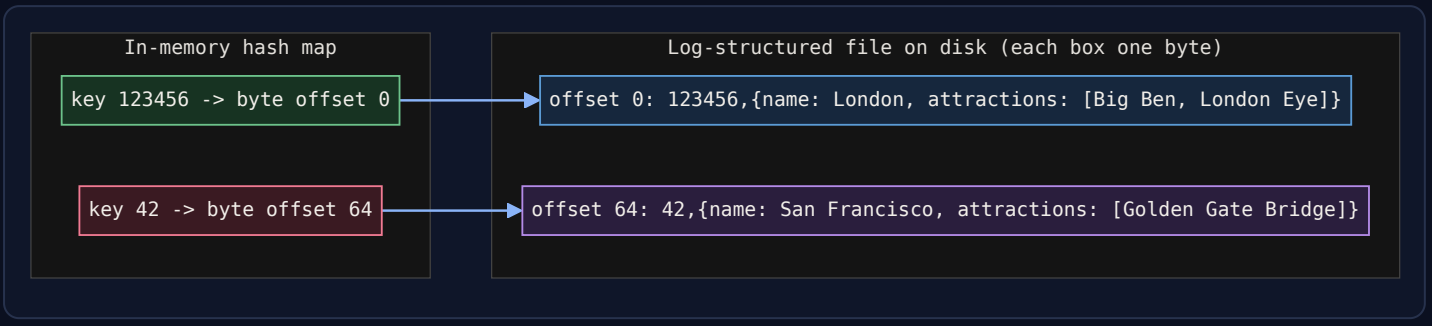
1. An index is **derived** from the primary data; adding or dropping one changes only query speed.
2. Every index must be updated on every write, so **indexes slow down writes** (nothing beats a plain append).
3. The core **trade-off**: indexes speed reads but cost writes, so databases do not index everything; the developer or DBA chooses based on actual query patterns.

Hash Indexes

What it is. An **in-memory hash map** (roughly constant-time lookup) mapping each key to the **byte offset** of its latest value in the data file.

How it works: **write** — append `key,value`, then set `hashmap[key] = offset` (new keys and updates alike); **read** — look up the offset, seek there, read the value.

Figure 3-1 (redrawn): an append-only CSV-like log of key-value records, with an in-memory hash map pointing at each key's byte offset



Real system: essentially **Bitcask**, the default storage engine of **Riak**. High-performance reads and writes **as long as all keys fit in RAM**; values may exceed memory, since a read costs one disk seek (or none if that part of the file is in the OS filesystem cache).

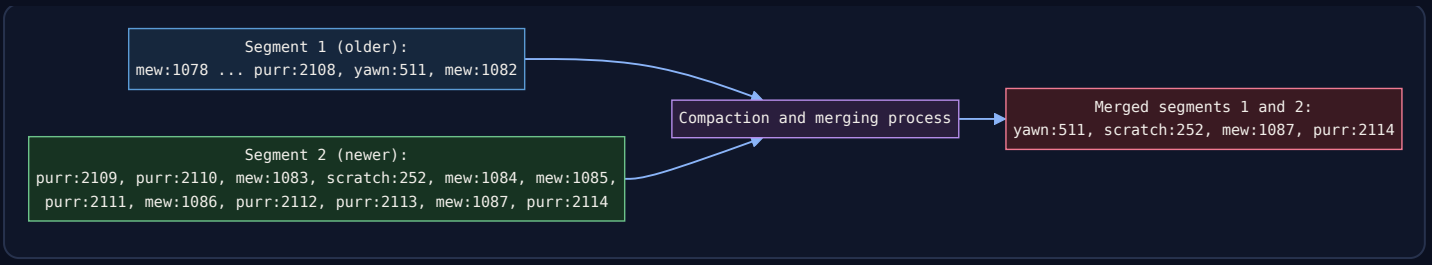
Good fit: few distinct keys, many updates each. Book example: key = cat-video URL, value = play count incremented on every play.

Problem: the log grows forever. Split it into **segments** (close a file at a certain size, start a new one); run **compaction** on closed segments (drop older duplicates, keep the latest value per key); and since compaction shrinks segments, **merge** several while compacting.

Figure 3-2 (redrawn): compacting one segment of a cat-video play-count log keeps only the newest value per key



Figure 3-3 (redrawn): compaction and merging of two segments done together, producing one new segment



Merging details: segments are **never modified**; the result goes to a **new file**, built by a **background thread** while reads and writes continue on old segments; then reads switch over and old files are deleted. **Each segment has its own hash map**; reads check newest to oldest, and merging keeps the count small.

Practical issues in a real implementation:

Issue	How it's handled
File format	Binary (length in bytes, then raw bytes) instead of CSV: faster, no escaping.
Deleting records	Append a **tombstone**; merging then drops all earlier values for that key.
Crash recovery	In-memory maps vanish on restart; rebuilding by reading whole segments is slow, so Bitcask stores a **snapshot of each segment's hash map** on disk to load quickly.
Partially written records	Bitcask uses **checksums** to detect and ignore corrupt tails after a crash.
Concurrency control	Writes are strictly sequential, so typically **one writer thread**; immutable segments allow **many concurrent readers**.

Why append-only instead of overwriting in place?

- Appends and merges are **sequential writes**, much faster than random writes on HDDs and, to a lesser degree, SSDs.
- **Concurrency and crash recovery are simpler** with immutable files: no value left half old, half new by a crash mid-overwrite.
- **Merging avoids fragmentation** over time.

Limitations:

- **The hash table must fit in memory**; on-disk hash maps perform poorly (random I/O, costly growth, fiddly collision handling).
- **Range queries are inefficient**: scanning `kitty00000` – `kitty99999` requires looking up each key individually.

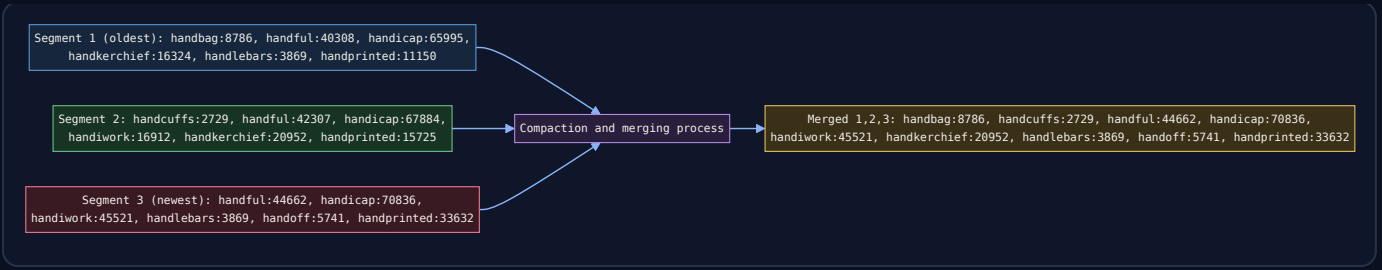
SSTables and LSM-Trees

The change. Require each segment to be **sorted by key**, with each key **once** per merged segment: a **Sorted String Table (SSTable)**.

Three advantages over hash-indexed log segments:

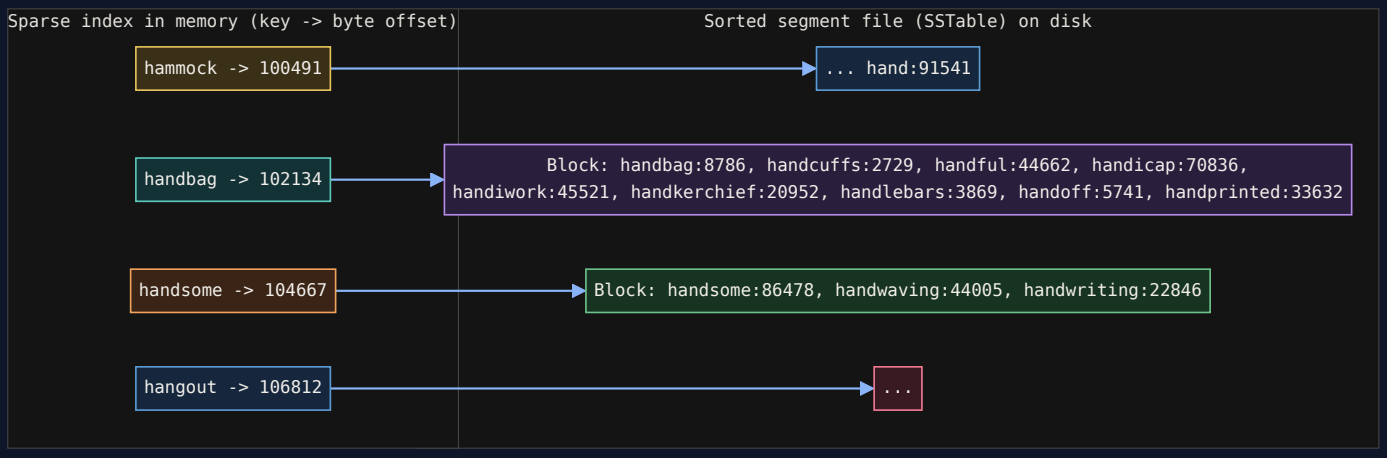
1. **Merging is simple and efficient even for files bigger than memory**, like mergesort's merge step: read the inputs side by side, repeatedly emit the lowest current key; the output is sorted. For a key in several inputs, keep the **newest segment's** value (each segment covers a time period, assuming adjacent segments are merged).

   Figure 3-4 (redrawn): merging three SSTable segments mergesort-style; for a key found in several segments, the value from the newest segment wins



2. **Not every key needs to be in memory.** A **sparse** index suffices: to find `handiwork` knowing only the offsets of `handbag` and `handsome`, jump to `handbag` and scan forward. One entry per **few kilobytes** is enough because scanning a few KB is fast. (With fixed-size records a binary search would need no index at all, but records vary in length.)
3. **Blocks can be compressed.** Group neighbouring records into a block and compress it before writing; each sparse-index entry points to a block start. Saves disk space *and* I/O bandwidth.

Figure 3-5 (redrawn): an SSTable with a sparse in-memory index; each entry points to the start of a compressible block of the sorted file



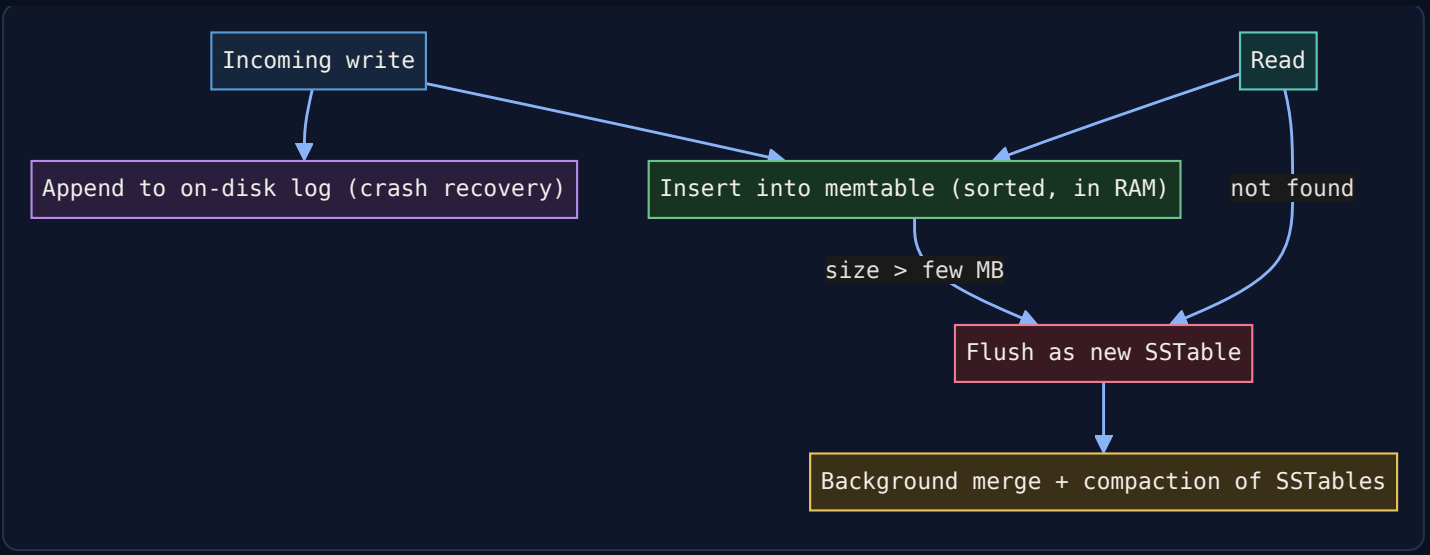
Constructing and maintaining SSTables

Writes arrive in random key order. A sorted structure **on disk** is possible (B-tree), but **in memory** it is easy with a self-balancing tree such as a **red-black tree** or **AVL tree**: insert in any order, read back sorted.

How the engine works:

1. **Write:** insert into the in-memory balanced tree, the **memtable**.
2. **Flush:** when it exceeds a threshold (typically a **few megabytes**), write it out as a new SSTable (already sorted), which becomes the newest segment; new writes go to a fresh memtable meanwhile.
3. **Read:** memtable, then newest segment, then older ones.
4. **Background:** merge and compact segments, dropping overwritten and deleted values.

The crash problem. A crash loses the memtable. Fix: also append every write to a separate **unsorted log** on disk, used only to rebuild the memtable after a crash and discarded once that memtable is flushed.



Making an LSM-tree out of SSTables

- This is essentially **LevelDB** and **RocksDB**, key-value engine libraries embedded into other applications; LevelDB can replace Bitcask in **Riak**.
- Similar engines power **Cassandra** and **HBase**, inspired by Google's **Bigtable** paper, which coined *SSTable* and *memtable*.
- First described by **Patrick O'Neil et al.** as the **Log-Structured Merge-Tree (LSM-Tree)**, building on log-structured filesystems; engines that merge and compact sorted files are **LSM storage engines**.
- **Lucene** (behind **Elasticsearch** and **Solr**) uses a similar approach for its **term dictionary**: a full-text index maps each word (**term**) to the IDs of documents containing it (the **postings list**), kept in SSTable-like sorted files merged in the background.

Performance optimizations

- **Missing-key lookups are slow** (memtable plus every segment back to the oldest, possibly disk reads each). **Bloom filters** approximate set contents and say for certain a key is *absent*, saving those reads.
- **Compaction strategies:**

Strategy	How it works	Used by
Size-tiered	Newer, smaller SSTables are successively merged into older, larger ones.	HBase; Cassandra (supports both)
Leveled	Key range split into smaller SSTables; older data moves into separate "levels"; more incremental, less disk space.	LevelDB (hence the name), RocksDB; Cassandra

Why LSM-trees work well: a simple, powerful idea (a cascade of SSTables tidied by background merging) that works with data far larger than memory, supports **range queries** (data is sorted), and gives **very high write throughput** (sequential disk writes).

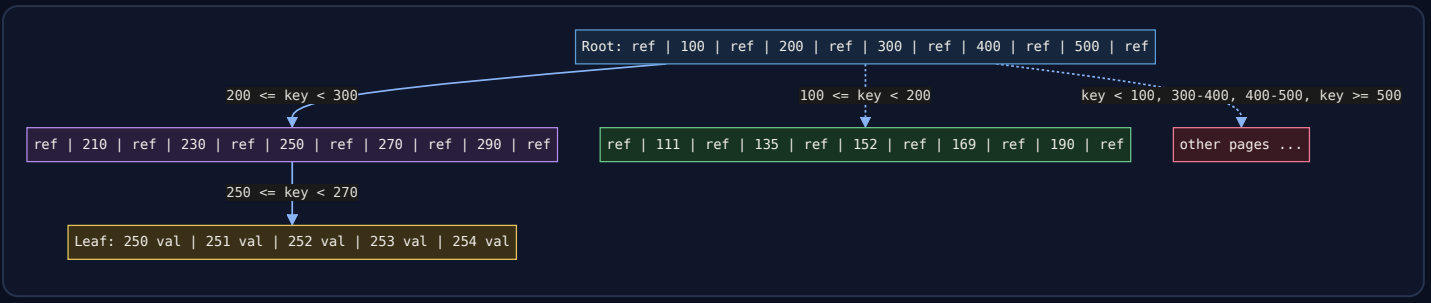
B-Trees

What it is. The **most widely used** index: introduced in **1970**, called "ubiquitous" less than 10 years later, still standard in almost all relational and many non-relational databases. Like SSTables, keys are sorted (efficient lookups and range queries), but the design philosophy differs:

	Log-structured (LSM)	B-tree
Unit of storage	Variable-size segments (several MB+)	Fixed-size **pages** (traditionally **4 KB**, sometimes bigger)
Write style	Always write a whole segment sequentially	Read/write one page at a time, overwriting in place
Hardware match	—	Close to how disks are organised (fixed-size blocks)

How it works: each page has an address, so pages reference each other like on-disk pointers, forming a **tree**. Lookups start at the **root**. A page holds keys interleaved with child references; each child covers a **continuous key range** bounded by the keys around its reference. Follow the matching reference down to a **leaf page**, which holds the keys with values inline or references to them. Book example: for key **251**, follow the root's reference between **200** and **300**, and keep descending.

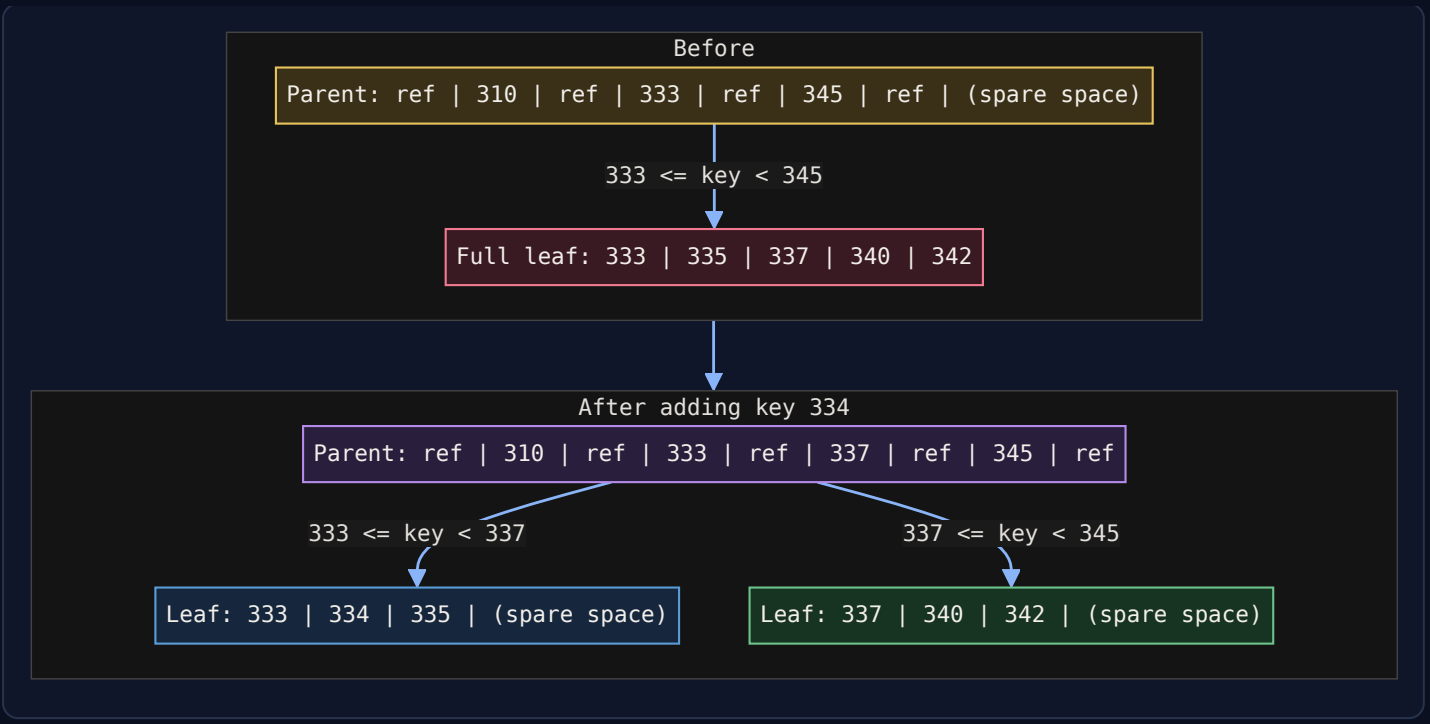
Figure 3-6 (redrawn): looking up user_id 251 by descending from the root through range boundaries to a leaf page



Branching factor: six in the figure, **several hundred** in practice (depends on the space references and boundaries take).

Updating and inserting: to **update**, find the leaf, change the value, write the page back (references stay valid). To **insert**, add the key to the page covering its range; if **full**, **split** it into two half-full pages and update the **parent**.

Figure 3-7 (redrawn): adding key 334 to a full leaf splits it into two pages with spare space; the parent gains the new boundary key 337



- The tree stays **balanced**: depth **O(log n)**; most databases fit in **three or four levels**.
- **Numbers to remember: 4 KB pages**, **branching factor 500**, **four levels** → up to **256 TB**.
- Insertion is intuitive; deletion while keeping balance is more involved.

Making B-trees reliable

- The basic B-tree write **overwrites a page in place**, assuming its location never changes — the opposite of LSM-trees, which only append and delete whole files. On an HDD that means moving the head, waiting for the platter to rotate, and rewriting the sector; an SSD must erase and rewrite a fairly large block of its chip.
- **Multi-page operations are dangerous**: a split writes two children *and* the parent; a crash halfway can corrupt the index (e.g. an **orphan page** with no parent).
- **Fix — the write-ahead log (WAL, a.k.a. redo log):** every change is appended to it **before** touching tree pages; after a crash it is replayed to restore a consistent tree.
- **Concurrency is harder:** threads could see the tree mid-change, so B-trees use **latches**. Log-structured engines merge in the background and swap segments atomically, without disturbing queries.

B-tree optimizations

- **Copy-on-write (append-only B-trees)** instead of overwrite + WAL (e.g. **LMDB**): write the modified page elsewhere and create new parent versions pointing to it. Also helps concurrency control (snapshot isolation, Chapter 7).
- **Abbreviated keys:** interior pages store only enough of a key to separate ranges → higher branching factor, fewer levels (a **B+ tree**, though the optimisation is so common it is rarely singled out).
- **Sequential leaf layout:** pages can be anywhere, so range scans may seek per page; many implementations try to lay leaves out in key order, hard to maintain as the tree grows. LSM-trees do this naturally by rewriting large chunks when merging.
- **Sibling pointers:** each leaf links to its neighbours, so ordered scans need not go back up to parents.
- **Fractal trees:** B-tree variants borrowing log-structured ideas to reduce seeks (nothing to do with fractals).

Comparing B-Trees and LSM-Trees

Rule of thumb: LSM-trees are usually **faster for writes**; B-trees are thought to **read faster** (LSM reads may check several structures and SSTables at different compaction stages). Benchmarks are often inconclusive and workload-dependent — test *your* workload.

Advantages of LSM-trees

- **Write amplification.** A B-tree writes data **at least twice** (WAL, then the page, plus more on splits), writes a **whole page** even for a few changed bytes, and some engines write a page **twice** to avoid a torn (partially written) page on power loss. LSM-trees also rewrite data repeatedly through compaction. One database write causing multiple disk writes over its lifetime is **write amplification**; it matters especially on **SSDs**, which wear out after a limited number of overwrites per block.
- **Higher write throughput.** When disk bandwidth is the bottleneck, every extra byte written costs throughput. LSM-trees often sustain more writes because they *sometimes* have lower write amplification (configuration- and workload-dependent) and write compact SSTables **sequentially** rather than overwriting scattered pages — most important on magnetic disks.
- **Better compression, smaller files.** B-trees waste space through **fragmentation** (after splits, or rows not fitting a page); LSM-trees periodically rewrite SSTables and have lower overhead, **especially with leveled compaction**.
- **On SSDs**, firmware often internally uses a log-structured algorithm to turn random writes sequential, so the engine's pattern matters less — but lower write amplification and less fragmentation still let more requests fit in the I/O bandwidth.

Downsides of LSM-trees

- **Compaction can interfere with live traffic:** requests may wait behind expensive compaction. Average throughput and response time barely suffer, but **high-percentile (tail) latencies** can be quite high. **B-trees are more predictable.**
- **Compaction can fall behind at high write rates.** Disk bandwidth is shared between initial writes (logging, memtable flushes) and compaction, which needs more as data grows. If it cannot keep up, **unmerged segments pile up** until the disk fills, and reads slow as they check more files. SSTable engines usually **don't throttle** incoming writes, so **monitor explicitly**.
- In a B-tree **each key lives in exactly one place**; an LSM engine may hold copies in several segments. So B-trees suit **strong transactional semantics**: many relational databases implement isolation with **locks on key ranges** attached directly to the B-tree (Chapter 7).

Aspect	B-tree	LSM-tree
Typical strength	Reads, predictable latency	Writes, throughput
Write amplification	WAL + full page writes (sometimes twice)	Repeated compaction (sometimes lower overall, depending on configuration and workload)
Disk pattern	Random page overwrites	Sequential segment writes
Space usage	Fragmentation leaves empty space	More compact, better compression
Background work	No background compaction	Compaction may hurt tail latency / fall behind
Key location	Exactly one place (good for range locks)	Possibly many segments
Maturity	Very mature, ingrained in databases	Increasingly popular in new datastores

Conclusion: B-trees are deeply embedded and consistently good, so they won't disappear; log-structured indexes are increasingly popular in new systems. No quick rule — test empirically.

Other Indexing Structures

So far: **key-value indexes**, like a **primary key** index (uniquely identifying a row, document, or vertex; other records refer to it by that key).

Secondary indexes are very common (`CREATE INDEX`), often essential for efficient joins:

```sql
-- A secondary index on a non-unique column
CREATE INDEX idx_orders_customer ON orders (customer_id);
-- Now this query can use the index instead of scanning every order:
SELECT * FROM orders WHERE customer_id = 812;
```

The difference from a primary index: **keys are not unique**. Fix by making each value a **list of matching row IDs** (like a postings list), or making keys unique by **appending the row ID**. Either way, secondary indexes can be **B-trees or log-structured**.

Storing values within the index

An index entry's value is either **the row itself** or a **reference** to it.

- **Heap file:** rows stored in no particular order (append-only, or reusing deleted space). With several secondary indexes, data lives in **one place** and each index points to it. Updating a value **without changing the key** can overwrite in place if not larger; if it **grows** and moves, update **all indexes** or leave a **forwarding pointer**.
- **Clustered index:** when the extra hop to the heap hurts reads, store the **whole row in the index**. **MySQL InnoDB**: the primary key is always clustered, and secondary indexes refer to the **primary key**, not a heap location. **SQL Server**: **one** clustered index per table.
- **Covering index / index with included columns:** store **some** columns in the index so certain queries are answered from it alone (the index **covers** the query):

```sql
-- SQL Server-style covering index
CREATE INDEX idx_emp_dept ON employees (dept_id) INCLUDE (salary);
SELECT salary FROM employees WHERE dept_id = 4;  -- answered from the index alone
```

Trade-off: clustered and covering indexes duplicate data → faster reads but more storage, write overhead, and effort to keep transactional guarantees (no visible inconsistency from the duplication).

Option	What the index stores	Read cost	Write/storage cost
Nonclustered (heap)	Pointer to row	Extra hop	Lowest
Covering	Some columns	No hop for covered queries	Medium
Clustered	Entire row	No hop	Highest

Multi-column indexes

- **Concatenated index** (most common): several columns joined into one key in fixed order. Like a **phone book** sorted by (last name, first name): finds a last name or last+first, but is **useless for first name alone**.
- **Multi-dimensional indexes:** important for **geospatial** data. A restaurant map needs everything inside the visible rectangle — a **2D range query**:

```sql
SELECT * FROM cafes
WHERE lat BETWEEN 48.850 AND 48.862      -- north-south band
  AND lon BETWEEN 2.330  AND 2.355;      -- east-west band
```

A normal B-tree or LSM index gives all rows in the latitude band (any longitude) *or* the longitude band, not both efficiently. **Solutions:** map 2D points to one number with a **space-filling curve** and use a B-tree; or, more commonly, a spatial index like an **R-tree** — **PostGIS** builds them on PostgreSQL's **Generalized Search Tree (GiST)**.

Not just maps: a 3D index on **(red, green, blue)** for colour search in e-commerce; **(date, temperature)** for weather readings in **2013** between **25 and 30 °C** without scanning all of 2013 (or vice versa). **HyperDex** uses this technique.

Full-text search and fuzzy indexes

The indexes above need **exact** keys or ranges; **fuzzy** querying finds **similar** keys (e.g. misspellings).

- Full-text engines expand to **synonyms**, ignore **grammatical variations**, find words **near each other**, and more.
- **Lucene** finds words within a given **edit distance** (1 = one letter added, removed or replaced).
- Lucene's term dictionary is SSTable-like with a small in-memory index: where **LevelDB** uses a sparse key set, **Lucene** uses a **finite state automaton** over the keys' characters (like a **trie**), which can be turned into a **Levenshtein automaton** to search efficiently within an edit distance.
- Other fuzzy techniques lean toward document classification and machine learning (see information-retrieval texts).

Keeping everything in memory

We tolerate awkward disk layouts because disks are **durable** and **cheaper per GB** than RAM. As RAM gets cheaper, many datasets fit in memory (possibly across machines): **in-memory databases**.

- **Cache-only** stores like **Memcached** accept losing data on restart.
- **Durable** ones use battery-backed RAM, a **change log** on disk, **periodic snapshots**, or **replication** (Chapter 5), reloading from disk or a replica on restart. Disk is only an append-only durability log — **all reads come from RAM** — and disk files also ease backups and external analysis.

Named systems: VoltDB, MemSQL, Oracle TimesTen (in-memory relational; vendors claim big speedups from dropping on-disk structure overhead); **RAMCloud** (open-source durable in-memory key-value store, log-structured in RAM and on disk); **Redis, Couchbase** (weak durability: write to disk **asynchronously**, so a crash can lose recent writes).

> **Common confusion:** in-memory databases are *not* fast mainly because they skip disk reads — a disk-based engine with enough RAM may never read disk either, thanks to the OS cache. The real win is avoiding **encoding in-memory structures into a disk-friendly format**.

Richer data models: structures hard to build on disk, e.g. **Redis** priority queues and sets, with a comparatively simple implementation.

Anti-caching: research suggests in-memory systems can exceed RAM by evicting **least recently used** (LRU) records to disk and reloading on access — like OS virtual memory/swap, but per **individual record** rather than whole pages. Indexes must still fit in memory (like Bitcask).

Future: non-volatile memory (NVM) may require further engine design changes; an active research area.

Transaction Processing or Analytics?

Early database writes usually matched a **commercial transaction** (a sale, supplier order, salary payment). The term **transaction** stuck, now meaning a group of reads and writes forming one logical unit.

> **Note:** here a transaction need not be ACID (Chapter 7). *Transaction processing* just means **low-latency** reads and writes, versus periodic **batch processing** jobs (Chapter 10).

OLTP. Blog comments, game actions, contacts — all look like business transactions: find a few records by key via an index; insert or update based on user input. Interactive, hence **online transaction processing (OLTP)**.

OLAP. Analytics is different: scan a **huge number of records**, read **a few columns** each, compute **aggregates** (count, sum, average) rather than return raw rows. E.g. total revenue per store in January? Banana sales during the latest promotion vs normal? Which baby-food brand is most often bought with brand X diapers? Such queries feed **business intelligence** reports: **online analytic processing (OLAP)** (the "online" is vague; probably means interactive exploratory queries).

Table 3-1 (paraphrased): OLTP vs OLAP characteristics

Property	OLTP	OLAP
Main read pattern	Each query fetches a few records by key	Aggregates computed across very many records
Main write pattern	Low-latency, random-access writes driven by user input	Event stream or bulk import (ETL)
Primarily used by	Customers/end users, through a web app	Internal analysts supporting decisions
What data represents	Latest state (current point in time)	History of events over time
Dataset size	GB up to TB	TB up to PB

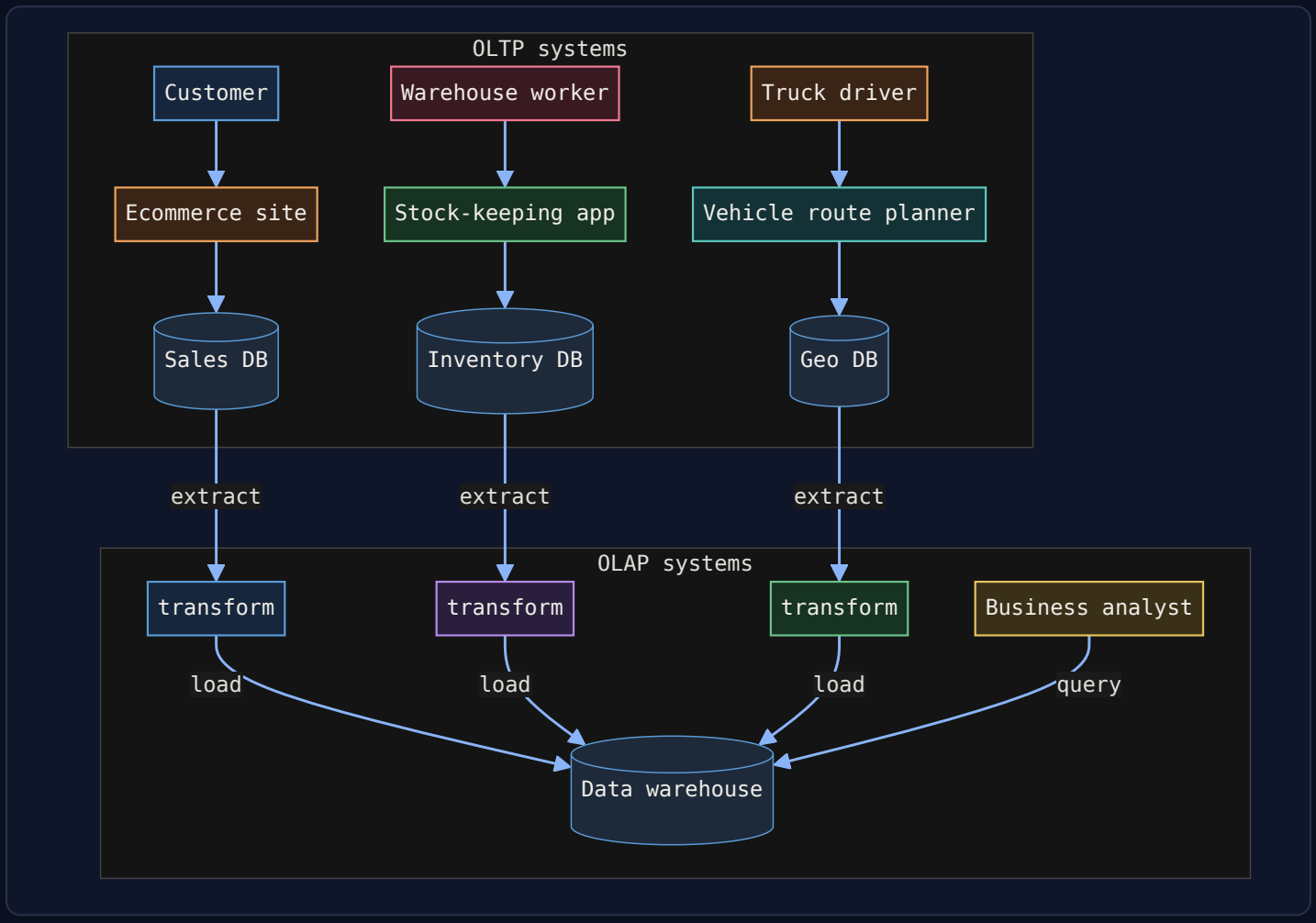
SQL handles both, and at first the same databases served both; from the **late 1980s and early 1990s** companies moved analytics to a **separate** database: the **data warehouse**.

Data Warehousing

A large company may run **dozens** of OLTP systems (website, point-of-sale, inventory, route planning, supplier management, HR…), each complex, with its own team. They must be **highly available** and **low-latency**, so DBAs resist analysts running expensive **ad hoc** scans on them.

A **data warehouse** is a separate database holding a **read-only copy** of data from all OLTP systems, which analysts can query freely. Data arrives via **Extract–Transform–Load (ETL)**: **extract** from OLTP databases (periodic dump or continuous update stream), **transform** into an analysis-friendly, cleaned schema, **load** into the warehouse.

Figure 3-8 (redrawn): simplified ETL. Each OLTP system is extracted and transformed separately, then loaded into one data warehouse that business analysts query



- Common in **large** enterprises, rare in **small** ones (few OLTP systems, little data — a normal SQL database or spreadsheet suffices).
- **Big advantage:** optimised for analytic access; the OLTP indexes above are **poor at analytic queries**.

The divergence between OLTP databases and data warehouses

- Warehouses are usually **relational** (SQL suits analytics); graphical tools generate SQL, visualise results, and support **drill-down** (from a total to its details) and **slicing and dicing** (cutting by different attributes).
- Both speak SQL, but **internals differ greatly**; vendors increasingly specialise in one. **Microsoft SQL Server** and **SAP HANA** support both, increasingly as two engines behind one SQL interface.
- **Commercial warehouses:** Teradata, Vertica, SAP HANA, ParAccel (expensive licences); **Amazon RedShift** is hosted ParAccel.
- **Open-source SQL-on-Hadoop** (younger): Spark SQL, Apache Hive, Apache Drill, Apache Tajo, Facebook Presto, Cloudera Impala; some build on Google's **Dremel**.

Stars and Snowflakes: Schemas for Analytics

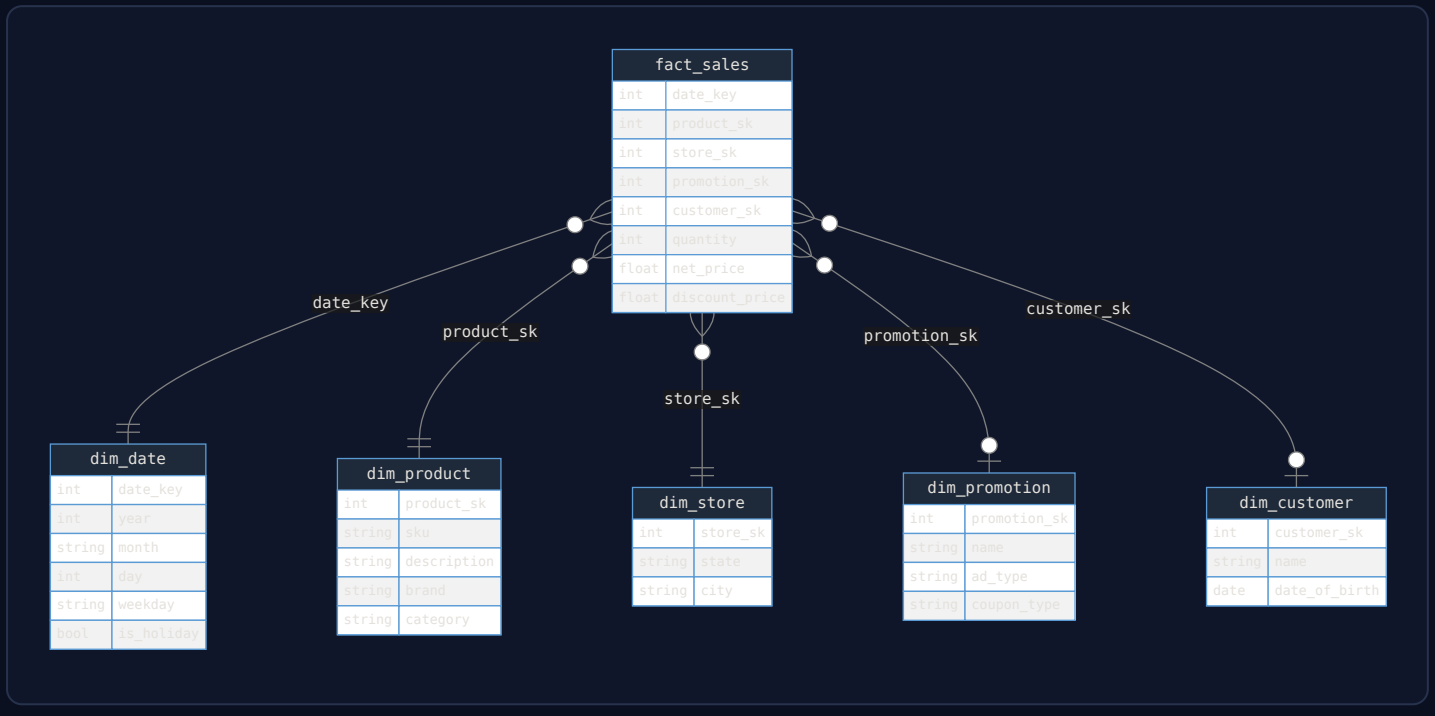
Analytics uses far fewer data models than OLTP; most warehouses use a **star schema** (**dimensional modeling**).

Fact table (e.g. `fact_sales` for a grocery chain): each row is an **event** at a particular time (here, a customer buying a product; in web analytics, a page view or click).

- Recording individual events maximises flexibility, but fact tables become **enormous**: **Apple, Walmart, eBay** may hold **tens of petabytes**, mostly facts.
- Some columns are **attributes** (sale price, supplier cost → profit margin); others are **foreign keys** to **dimension tables** describing the event's **who, what, where, when, how and why**.

Dimension tables: e.g. `dim_product`, one row per product type (SKU, description, brand, category, fat content, package size…); each fact row references the product sold (several products in one purchase = several fact rows). Even **date and time** get a dimension so extra info like **public holidays** can be attached and compared.

Figure 3-9 (redrawn): star schema for a grocery retailer. fact_sales sits in the centre; its foreign keys point at five dimension tables (sample rows in the book: product 31 = Fish food by Aquatech, store 3 = Palo Alto CA, promotion 19 = Aquarium deal, customer 191 = Bob, date 140102 = Thu 2 Jan 2014)



- **Why "star":** the fact table in the middle with dimensions around it like rays.
- **Snowflake schema:** dimensions split into sub-dimensions (e.g. separate `brand` and `category` tables referenced by `dim_product`): more **normalised**, but stars are usually preferred as **simpler for**

analysts.

- **Tables are wide:** fact tables often exceed **100 columns**, sometimes several hundred; dimensions too (e.g. `dim_store` with services offered, in-store bakery, floor area, opening date, last remodel, distance to highway).

Column-Oriented Storage

Fact tables can have **trillions of rows** and **petabytes**; dimension tables are much smaller (**millions** of rows), so focus on **facts**.

Key observation: a typical warehouse query touches only **4 or 5** of 100+ columns (`SELECT *` is rare). The book's Example 3-1 asks whether people prefer fresh fruit or candy by weekday across all of 2013, touching only `date_key`, `product_sk` and `quantity`. A similar query of ours:

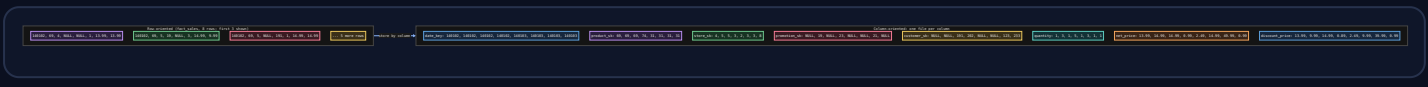
```sql
SELECT d.month, s.region, SUM(f.net_price) AS revenue   -- only a few columns needed
FROM fact_sales f
  JOIN dim_date  d ON f.date_key = d.date_key            -- attach date details
  JOIN dim_store s ON f.store_sk = s.store_sk            -- attach store details
WHERE d.year = 2015 AND s.region IN ('North', 'West')    -- filter
GROUP BY d.month, s.region;                              -- one total per month per region
```

It reads just `date_key`, `store_sk` and `net_price` from `fact_sales`, but across a huge number of rows.

Why row storage struggles: OLTP databases are **row-oriented** (a row's values adjacent; document databases similarly). Even with indexes on `date_key` or `product_sk`, a row store loads whole 100+-attribute rows from disk, parses and filters them — slow.

The column-oriented idea: store each **column** together in its **own file**; queries read and parse only the columns they need. (Works for non-relational data too: **Parquet** is a columnar format supporting a document model, based on Google's **Dremel**.)

Figure 3-10 (redrawn): the book's eight fact_sales rows stored as one file per column; position k in every file belongs to row k



Crucial rule: every column file lists rows in the **same order**; row 23 = the 23rd entry of each file.

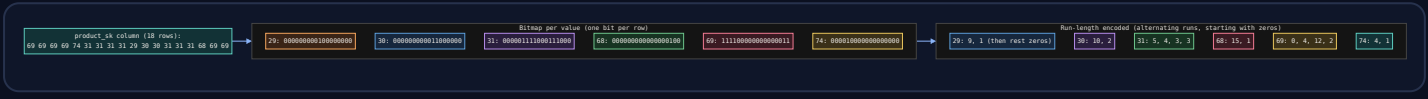
Column Compression

Columns are often repetitive, so **compression** cuts I/O further.

Bitmap encoding (especially effective in warehouses):

1. Columns usually have few **distinct values** relative to rows (**billions** of sales, **100,000** products).
2. For **n distinct values**, make **n bitmaps**, **one bit per row** (1 if the row has that value).
3. Very small n (e.g. ~**200** countries): store bitmaps as-is.
4. Larger n: bitmaps are mostly zeros (**sparse**), so add **run-length encoding** (lengths of alternating runs of 0s and 1s).

Figure 3-11 (redrawn): an 18-row product_sk column turned into one bitmap per distinct value, then run-length encoded



Bitmaps make common filters cheap: `WHERE product_sk IN (30, 68, 69)` → bitwise **OR** of three bitmaps; `product_sk = 31 AND store_sk = 3` → bitwise **AND** of two bitmaps (works because bit k is the same row in every column). Many other compression schemes exist.

Column-oriented storage vs column families (common confusion): **Cassandra** and **HBase** have *column families* from **Bigtable**, but within a family they store all of a row's columns together with a row key and use no column compression. The Bigtable model is mostly **row-oriented**.

Memory bandwidth and vectorized processing

Besides disk-to-memory bandwidth, analytic engines care about moving data from **RAM into the CPU cache**, avoiding **branch mispredictions** and **bubbles** in the CPU pipeline, and using **SIMD** (single-instruction-multi-data) instructions. Column layouts help: the engine takes a chunk of compressed column data that fits in the **L1 cache** and loops over it tightly with **no function calls** — far faster than per-record code full of calls and branches. Compression fits **more rows** in L1, and operators like bitwise AND/OR can run **directly on compressed chunks**. This is **vectorized processing**.

Sort Order in Column Storage

Rows can stay in **insertion order** (insert = append to each column file), or be given a **sort order**, like SSTables, usable as an index.

- **Columns cannot be sorted independently** (you would lose which values form a row); sort **whole rows**.
- The DBA picks **sort keys** from common queries: if queries target date ranges, make `date_key` the **first** key so only that range is scanned. A **second** key orders ties, e.g. `date_key` then `product_sk` groups same-day sales of a product.

Sorting boosts compression: a first sort column with few distinct values forms **long runs**, so run-length encoding can shrink it to a **few kilobytes** even for **billions** of rows. The first key benefits most; later columns look nearly random. Still a net win.

Several different sort orders

From **C-Store**, adopted by **Vertica**: data is **replicated** to several machines anyway for fault tolerance, so store each replica **sorted differently** and let each query use the best-fitting one.

	Multiple sort orders (column store)	Multiple secondary indexes (row store)
Where the full data lives	Each sorted copy contains the column values themselves	Each row in one place (heap file or clustered index)
What the extra structures hold	No pointers — just columns of values	Pointers to matching rows

Writing to Column-Oriented Storage

Columnar storage, compression and sorting speed **reads** (the main warehouse load) but make **writes harder**: **update-in-place is impossible** with compressed columns, and inserting **midway** through a sorted table would likely mean **rewriting every column file** (rows are identified by position).

Solution: LSM-trees again. Writes go to a sorted **in-memory store** (row- or column-oriented, it doesn't matter); when enough accumulate, they are **merged** with on-disk column files into **new files in bulk** (essentially what **Vertica** does). **Queries** combine on-disk data with recent in-memory writes, and the **query optimizer hides** this, so inserts, updates and deletes appear immediately.

Aggregation: Data Cubes and Materialized Views

Not every warehouse is columnar, but column stores can be **much faster for ad hoc analytics** and are gaining popularity.

Materialized aggregates. Queries often use `COUNT`, `SUM`, `AVG`, `MIN`, `MAX`; if many reuse the same aggregates, **cache** them rather than recompute from raw data.

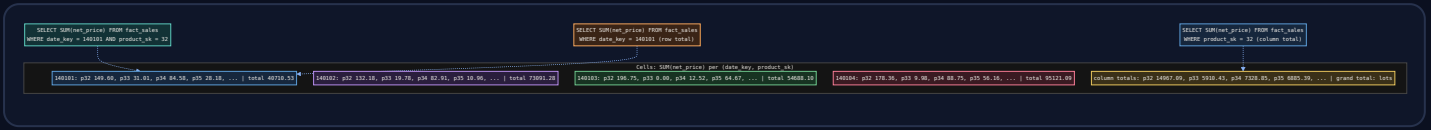
	Virtual view	Materialized view
What it is	A named query shortcut	The query's results really stored **on disk**
On read	SQL engine expands it into the underlying query on the fly	Read precomputed results directly
On data change	Nothing to do	Must be **updated** (denormalised copy); database can do it automatically
Cost	Slow reads for heavy queries	More expensive **writes**

Because of write cost, materialized views are rare in **OLTP** but more sensible in **read-heavy warehouses** (case-dependent).

```
-- PostgreSQL-style example
CREATE MATERIALIZED VIEW daily_store_revenue AS
  SELECT date_key, store_sk, SUM(net_price) AS revenue
  FROM fact_sales
  GROUP BY date_key, store_sk;          -- precompute totals per day per store
REFRESH MATERIALIZED VIEW daily_store_revenue;  -- recompute after new sales arrive
```

Data cube / OLAP cube: a materialized view that is a **grid of aggregates by several dimensions**. With two dimensions, **date** and **product**, each cell holds e.g. `SUM(net_price)` for that date–product pair; summing a row or column gives totals with one dimension removed (per date, or per product).

Figure 3-12 (redrawn): two dimensions of a data cube (date_key by product_sk). Each cell is SUM(net_price) for that pair; summing along a row gives the per-date total, along a column the per-product total



- More dimensions (Figure 3-9 has **five**: date, product, store, promotion, customer) make a **hypercube**, same idea.
- **Advantage:** some queries become near-instant (e.g. yesterday's sales per store = read totals along one dimension).
- **Disadvantage: less flexible** than raw data: e.g. the share of sales from items priced above **$100** is impossible if price isn't a dimension.
- So most warehouses **retain raw data** and use cubes only as a **speed-up for particular queries**.

Summary

- **Two categories** of storage engines: **OLTP** (user-facing, huge request volume, each query touches few records found by key via an index; **disk seek time** is often the bottleneck) and **analytics/OLAP** (business analysts, fewer queries each scanning millions of records; the limit is **disk bandwidth**, answered by **column-oriented storage**).
- **Two OLTP schools: log-structured** (only append and delete obsolete files; Bitcask, SSTables, LSM-trees, LevelDB, RocksDB, HBase, Cassandra, Lucene) and **update-in-place** (fixed-size overwritable pages; **B-trees**, in all major relational and many non-relational databases). Log-structured engines turn **random writes into sequential writes**, raising write throughput on HDDs and SSDs.
- **Hash indexes** (Bitcask): keys must fit in RAM, no range queries; segments + compaction + merging, tombstones, checksums.
- **SSTables/LSM-trees:** memtable → sorted SSTables → background merging (size-tiered or leveled); sparse index, compressed blocks, Bloom filters, crash log.
- **B-trees:** 4 KB pages, in-place updates, splits keep balance (4 levels × 500 branching × 4 KB = 256 TB); WAL and latches; copy-on-write (LMDB), abbreviated keys (B+ tree), sibling pointers, fractal trees.

- **LSM vs B-tree:** LSM usually better for writes and space; B-tree for reads, predictable latency, key-range locking. Benchmark your workload.
- **Other indexes:** secondary; heap vs clustered (InnoDB, SQL Server) vs covering; concatenated vs multi-dimensional (R-trees, PostGIS, HyperDex); fuzzy (Lucene automata). **In-memory DBs** win by avoiding encoding overhead, not disk reads.
- **Warehouses** (via **ETL**, **star**/snowflake schemas): indexes matter less than **very compact encoding** — columnar storage, bitmap/RLE compression, vectorised processing, sort orders (C-Store/Vertica), LSM-style writes, and **materialized views/data cubes** trading flexibility for speed.
- **Takeaway:** knowing engine internals helps you choose tools, predict the effect of tuning parameters, and read your database's documentation.

Interview angle

1. B-tree vs LSM-tree — which would you pick? *30-second answer:* B-trees update fixed-size pages in place with a WAL; each key lives in one place, reads are fast and latency predictable — the relational default. LSM-trees buffer writes in a memtable, flush sorted SSTables sequentially, and compact in the background — higher write throughput and better compression, but reads may check several files and compaction can hurt tail latency. Write-heavy (logs, metrics, time-series) → LSM; read-heavy/transactional → B-tree. Benchmark.

- *Follow-up: What is write amplification?* One logical write causing several physical writes over time (WAL + page writes in B-trees; repeated compaction in LSM). Limits throughput and wears out SSDs.
- *Follow-up: How does an LSM avoid checking every SSTable for a missing key?* Per-SSTable Bloom filters say "definitely not here".
- *Follow-up: What if compaction can't keep up?* Unmerged segments pile up, reads slow, disk can fill; engines usually don't throttle writes, so monitor.

2. How does an LSM-tree handle a write and survive a crash? *30-second answer:* The write is appended to an on-disk log and inserted into the sorted memtable; at a few MB the memtable is flushed as an immutable SSTable. After a crash the log rebuilds the memtable; after a flush it is discarded.

- *Follow-up: Deletes?* A tombstone; compaction drops older values.
- *Follow-up: Size-tiered vs leveled?* Size-tiered merges smaller/newer into bigger/older tables (HBase); leveled splits key ranges into small tables across levels, more incremental and less disk space (LevelDB, RocksDB). Cassandra supports both.

3. How does a B-tree stay balanced and crash-safe? *30-second answer:* An overflowing page splits into two half-full pages with a new boundary in the parent, keeping depth O(log n), usually 3–4 levels. Changes go to a write-ahead log first so a crash mid-split can be repaired; latches protect concurrent access.

- *Follow-up: Alternative to a WAL?* Copy-on-write pages (LMDB).

4. OLTP vs OLAP, and why a separate data warehouse? *30-second answer:* OLTP = many small key-based user reads/writes, latest state, GB–TB. OLAP = few large aggregate scans for analysts, event history, TB–PB. Analyst queries on OLTP databases hurt customer latency and OLTP indexes are poor for scans, so ETL copies data into a warehouse optimised for analytics (often columnar, star schema).

- *Follow-up: Star schema?* Central fact table of events with foreign keys to dimension tables; snowflake further normalises dimensions.

5. Why are column stores fast for analytics? *30-second answer:* Queries touch 4–5 of 100+ columns, so reading separate column files avoids most I/O. Columns compress very well (bitmaps, RLE), especially the sort key, and compressed chunks fit CPU cache for vectorised processing.

- *Follow-up: Writes?* Buffer in memory (LSM-style) and periodically merge into new column files; queries combine both (Vertica).
- *Follow-up: Are Cassandra/HBase column stores?* No — column families are row-oriented within each family, with no column compression.

6. Clustered vs covering vs heap-file indexes? *30-second answer:* Heap-file indexes point to rows stored elsewhere (no duplication, extra hop). Clustered indexes hold the full row (InnoDB primary key; one per

table in SQL Server). Covering indexes hold some columns so certain queries need only the index. More duplication = faster reads, more storage and write cost.

Self-check

1. **Why is appending to a log fast, and why is reading from an unindexed log slow?** Appending is a sequential write at the end of the file; reading without an index scans the whole file, O(n).
2. **What is the main limitation of Bitcask-style hash indexes, and what workload suits them?** All keys must fit in RAM and range queries are inefficient. Suits many updates to a modest number of keys (e.g. play counts per video URL).
3. **Why can an SSTable use a sparse index while a hash-indexed log cannot?** Keys are sorted, so neighbouring keys' offsets bound the range to scan; in an unsorted log a key could be anywhere.
4. **Trade-off: your service ingests millions of sensor readings per second and rarely updates old data. B-tree or LSM-tree? Why?** LSM-tree: sequential writes, higher write throughput, better compression. Watch compaction and tail latency.
5. **What happens in a B-tree when you insert a key into a full page, and why is that risky without a WAL?** The page splits and the parent is updated — several page writes. A crash in between could leave an orphan page; the WAL restores consistency.
6. **How big can a 4-level B-tree with 4 KB pages and branching factor 500 get?** Up to 256 TB.
7. **Scenario: a map app needs "all shops inside this rectangle". Why won't a concatenated (lat, lon) B-tree index work well, and what should you use?** It narrows by latitude but must then scan all longitudes in that band. Use a spatial index such as an R-tree (e.g. PostGIS on GiST) or a space-filling curve.
8. **What is the real reason in-memory databases are faster?** They avoid encoding in-memory structures into disk-friendly form — not just avoiding disk reads, which the OS cache already avoids for disk-based engines with enough RAM.
9. **In a column store, why can't each column be sorted independently? What benefit does sorting give?** Rows are reconstructed by position; independent sorting breaks alignment. Sorting whole rows speeds range filters on the sort key and creates long runs that compress extremely well.
10. **What's the downside of relying on a data cube instead of raw data?** Inflexibility: only questions along chosen dimensions (e.g. no share of sales for items over $100 if price isn't a dimension). Keep raw data; use cubes as a speed-up.